

NE555 Precision Timer IC

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1. Technical Specifications & Functional Blocks

The NE555 is a precision timing circuit capable of producing accurate time delays or oscillation. In the time-delay (monostable) mode, the timed interval is controlled by a single external resistor and capacitor network.

Specification	Standard Limit	Notes
Supply Voltage (VCC)	4.5V to 16.0V DC	Functional down to 4.5V; nominal 5V to 12V
Max Output Current	200 mA (Source / Sink)	Can drive relays and LEDs directly
Timing Range	Microseconds to Hours	Determined by external RC values
Threshold Voltage	$2/3 \cdot VCC$	Resets internal flip-flop
Trigger Voltage	$1/3 \cdot VCC$	Sets internal flip-flop (active low)

2. Pinout Configuration (8-Pin DIP & SOIC)

- Pin 1 (GND): Ground reference (0V).
- Pin 2 (TRIG): Trigger input. When voltage drops below $1/3 VCC$, comparator sets flip-flop and output goes HIGH.
- Pin 3 (OUT): Output pin. High state delivers $VCC - 1.5V$; low state pulls to $GND + 0.1V$.
- Pin 4 (RESET): Active-low master reset. Connect to VCC for normal operation. If left floating, noise triggers false resets.
- Pin 5 (CTRL): Control voltage access. Directly biases internal $2/3 VCC$ divider. Must connect 10 nF capacitor to GND for noise immunity.
- Pin 6 (THRES): Threshold input. When voltage rises above $2/3 VCC$, comparator resets flip-flop and output goes LOW.
- Pin 7 (DISCH): Open-collector discharge pin. Discharges timing capacitor to ground in tandem with output state.
- Pin 8 (VCC): Positive DC supply voltage (+5V to +15V).

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3. Astable Multivibrator (Oscillator) Design Formulas

In astable mode, the timer free-runs as a square wave generator with frequency: $f = 1.44 / ((R1 + 2 \cdot R2) \cdot C1)$ High Time $T_{high} = 0.693 \cdot (R1 + R2) \cdot C1$ Low Time $T_{low} = 0.693 \cdot R2 \cdot C1$ Duty Cycle $D = (R1 + R2) / (R1 + 2 \cdot R2)$

Design Rule: R1 must be at least 1 kOhm to prevent shorting VCC to ground during the discharge cycle through Pin 7.

4. Troubleshooting & Circuit Diagnostics

- Symptom 1: Timer does not oscillate; output remains stuck HIGH. Inspection: Check Pin 4 (RESET). If Pin 4 is floating or below 0.7V, the timer is held in reset. Check Pin 6 (Threshold); if voltage cannot reach 2/3 VCC due to capacitor leakage, output will never reset.
- Symptom 2: Severe supply rail noise whenever output switches. Cause: Large 'crowbar' current spike (up to 400 mA) occurs internally when output totem-pole switches states. Solution: Solder a 0.1 uF ceramic cap in parallel with a 10 uF tantalum cap between Pin 8 and Pin 1.
- Symptom 3: Erratic timing intervals under motor or relay loads. Solution: Add 10 nF capacitor from Pin 5 to GND to stabilize comparator reference voltage.